

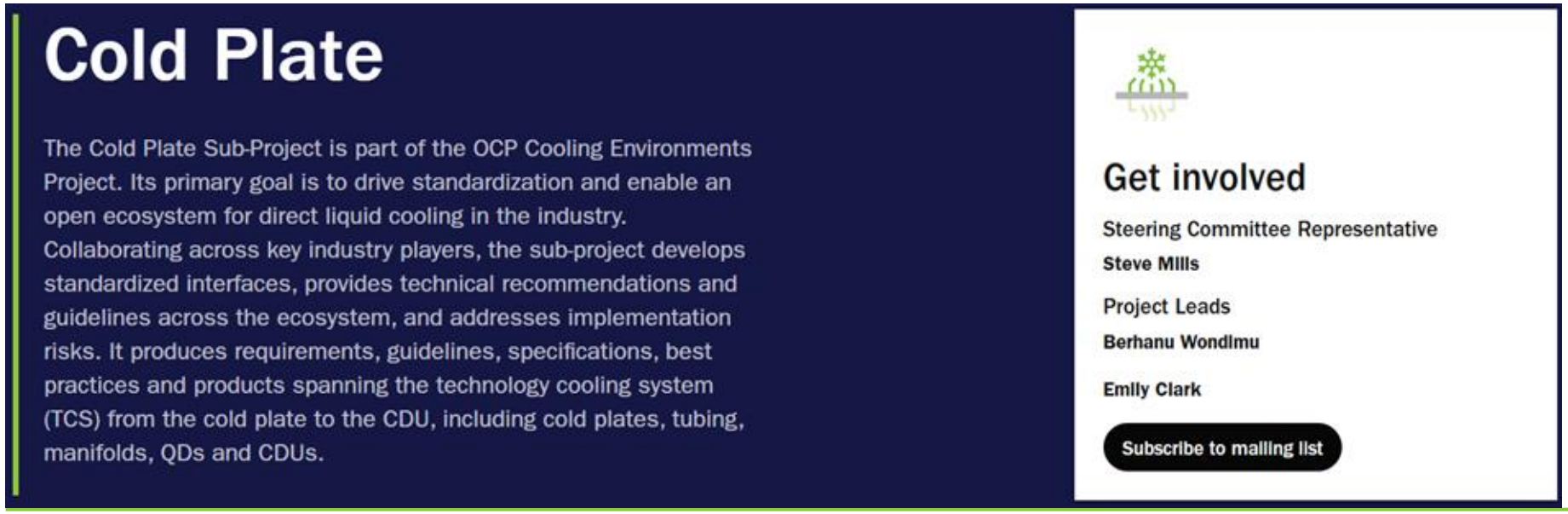
Cold Plate Sub-Project Overview

Berhanu Wondimu, *OCP Cold Plate Sub-project Co-lead, Intel*
Emily Clark, *OCP Cold Plate Sub-project Co-lead, Dell Technologies*



Cold Plate Workstream Lightning Talks

WORKSTREAM	PRESENTER
Cold Plate Sub-project Updates	Berhanu Wondimu
DLC E1.S	Anthony Constantine
Coolant Fluids Workstream Overview	Keegan Yaroch
Wetted Materials	Dr. Florian Rüping
PBMC Specification & Updates	Tim Marquis
UQDv2.0 Update	Elizabeth Langer



- Resources:
  - Subscription information
  - Workstream documentation (Google drive)
  - Community meeting links
  - wiki page

https://www.opencompute.org/projects/cooling-environments/cold-plate/

Cold Plate Subproject: Workstreams

WORKSTREAM	WORKSTREAM LEADS
DLC E1.S	Anthony Constantine, Brad Zakaib
Coolant Fluids Workstream	Philip Yu, Keegan Yaroch & Lilach Butchmits
Wetted Materials	Dr. Florian Rüping, Alex Pica
ORv3 BMQC/PBMC Update (CLA Group)	Vijay Mohan & Glenn Charest
Cold Plate Performance Workstream (Simulations and Experiments)	Thomas Rees, Andrea Francescon







- **Multiple CLA groups**

Cold Plate Subproject: Approved White Papers

- Guidelines for Using Propylene Glycol-Based Heat Transfer Fluids in Liquid Cooling Racks
- Pre-Commission Preparation of Technology Cooling System (TCS) Row Manifolds
- 2P Refrigerant-Based DLC Technology for Next Generation AI Clusters with High TDP Accelerators
- Guidelines for Implementing Passive TCS Using LHP
- Liquid to Liquid CDU Test Methodology and Performance Rating
- Guidelines for Using Dielectric Heat Transfer Fluids in Two- Phase Cold Plate-Based Liquid-Cooled Racks White Paper
- White Paper: Considerations for Shipping and Storage of 1-Phase Cold Plate Liquid Cooled Racks & Servers
- Large Quick Connector Specification (V2.0.0)

Cold Plate Subproject: Approved Specifications

- Single-Phase Direct-to-Chip Cold Plate Test Specification
- Universal Quick Disconnect (UQD) Specification V2.0.0
- Base Specification for Two-Phase Dielectric Fluid-Based Heat Transfer Fluid V1.0.0
- Base Specification for Single-Phase Propylene Glycol-Based Cold Plate Coolants (PG 25) V1.0.0
- Base Specification for Single Phase Water-Based Cold Plate Coolants
- Large Quick Connector Specification V2.0.0
- Pivoting Blind Mate Coupling (PBMC) Specification
- Rope Leak Sensor Base Specification V1.0.0

Cold Plate Subproject: Contributions in process

- **White Papers:**
  - Mini Quick Disconnect (Mini-QD) Design Specification
  - Thermal boundaries in AI infrastructure
  - Guideline for using water-based heat transfer fluid in single-phase cold plate-based liquid-cooled racks

- **Specifications:**
  - MQD/MQDB OCP Design Specification

Cold Plate Subproject: Invited Speakers

- Advanced TIM materials (Diamond ceramics) - Coherent
- Dissolved oxygen removing technology from water coolant - WATTS
- Application of Liquid-Cooled Load Banks (LCLBs) for CDU and Loop Validation - Avtron Power Solutions
- Metal-based TIM for high power Chips - Indium
- Two Phase Cooling for Data Centers 101 - BP
- Coolant Chemistry Sensor Interface and Telemetry - Nordran
- Rack-Scale Weight Reduction for Liquid-Cooled Server Systems Using Lightweight Materials - Wiwynn
- Technical Challenges and Key Considerations in Transitioning from Single-Phase to Two-Phase DLC - Wiwynn
- Elastomeric Seal Contamination in Liquid-Cooled Racks - Pacific Rubber

Cold Plate Subproject: Future Work

- Ingredient level enabling and modular design for interoperability
- Total technology cooling system enabling through standardization
- Testing/verification methodology of secondary loop components
- TIM material characterization
- Cross collaboration with other projects/subprojects
  - Telemetry & control system integration
  - Cold plate to the facility (heat reuse & data center facilities)
- Community driven standardization and knowledge sharing

Please join our mailing list to collaborate!

Call to Action

- Get Involved!
  - Seeking community driven topics and workstreams
  - Openly enable data center cooling
  - Standardize/define critical parameters and environmental conditions
  - Enable interoperability in the supply chain
- Join the mailing lists today:
  - Cooling Environments
  - Cold Plate Subproject



Monthly Community Call
(2nd Wednesday of each month)

COOLING:
COLD PLATE

DLC E1.S

Anthony Constantine

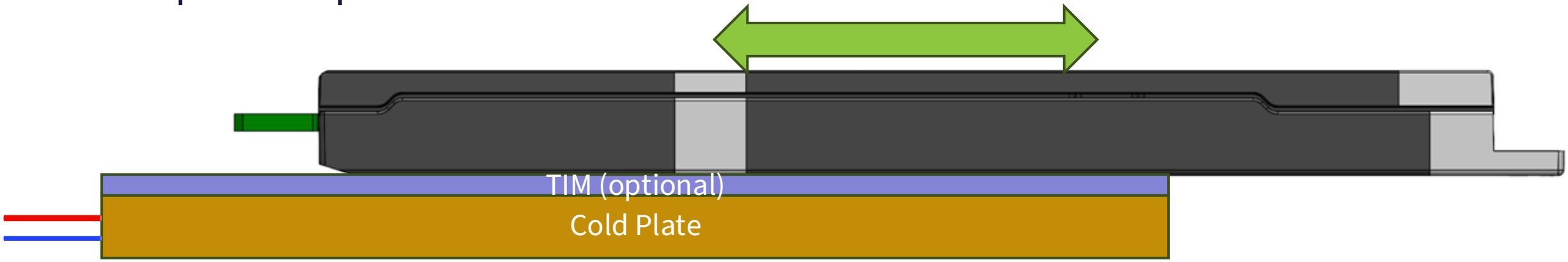
Distinguished Member of Technical Staff at Micron Technology

OCP BRONZE™ Member

Credit: Brad Zakaib @ CoolIT Systems

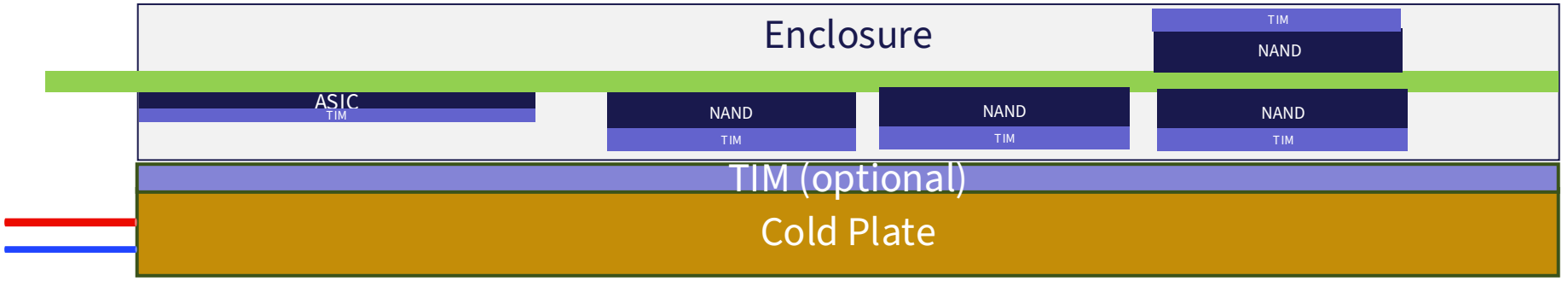
How do we Liquid Cool SSDs today? Hot Plug

- Serviceability in a MUST!
- Must be able to insert/remove an SSD while the host system is in operation
- Insertion/removal needs to be done by hand
  - using single sided currently to make this possible
- TIM may or may not exist
- Takes up more space = less SSDs



How do we Liquid Cool SSDs today? Thermals

- Thermal resistance is higher with current solution
  - Components are on both sides of device
  - Various components have different temperature limits
  - Various components have different heights
  - Hot plug means the TIM may not be at ideal pressure



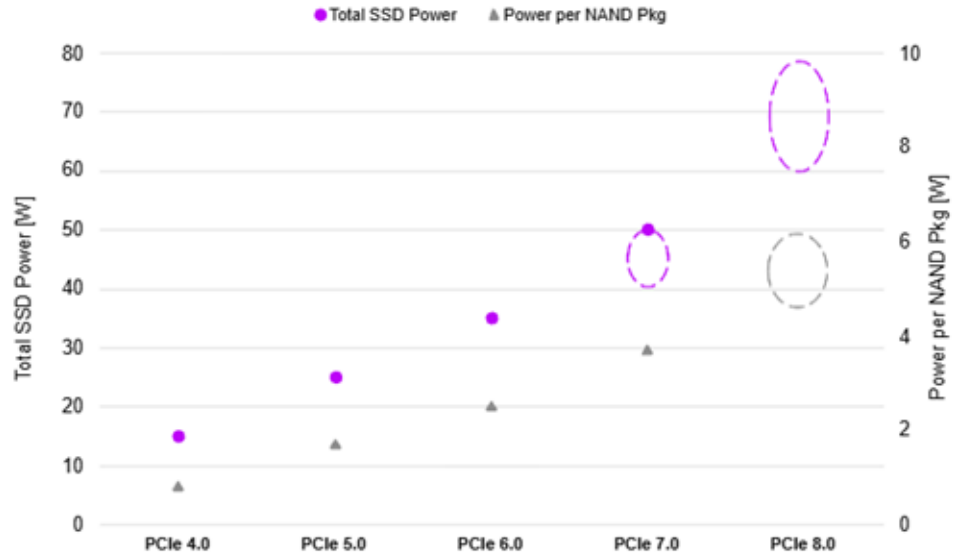
Do SSD's Care?

Short answer: Yes.

New developments on the block

PCIe roadmap & power trends

- SSD power has increased linearly from PCIe 4.0 → PCIe 6.0
- Sizeable jump projected for PCIe 7.0
- NAND power tracking total SSD power
- PCIe 7.0 at ~50 W could mean ~3.5 W per NAND package (in E1.S) — where does PCIe 8.0 land?

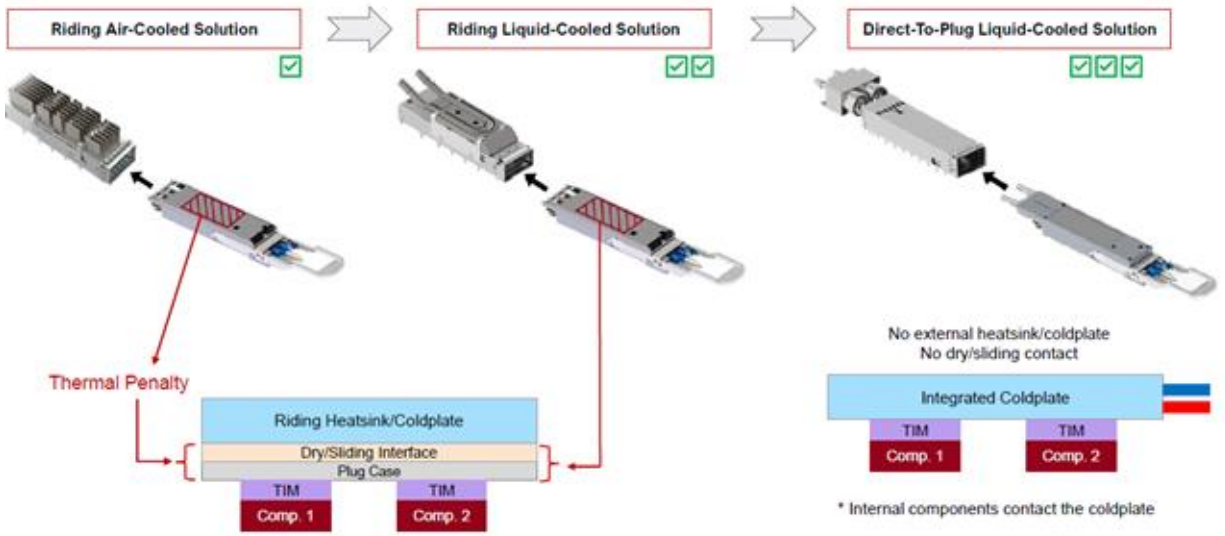


OCP GLOBAL SUMMIT 2026 Scaling Innovation for the AI Era.

How could we solve this problem?

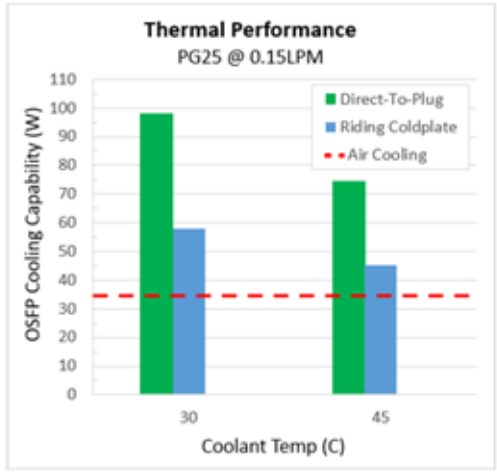
Ciena presentation from last year on OSFP

Thermal Management Evolution – Pluggable Optical Modules



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Thermal Performance



- Heat distribution: non-uniform, assuming ~60% power on DSP with localized hotspot.
- Riding coldplate: sliding TIM at the interface between plug and coldplate.

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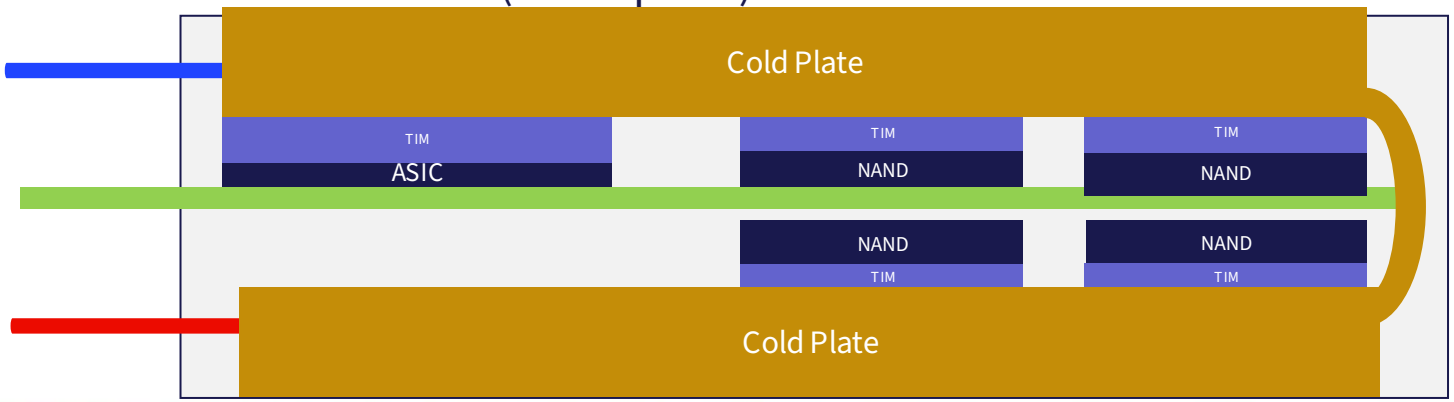
Workstream Started

- Brad Zakaib (CoolIT) began a DLC integrated SSD workstream
- Sponsors: CoolIT, Micron, Staubli
- The aim is to develop a reference design for a liquid cooling integrated E1.S SSD designed to enable maximum cooling performance and maintain electrical and thermal interoperability.

Concept with Integrated Cold plate

- Benefits
  - Lower thermal impedance
  - Better cooling capability
  - Possibly more compact (can place more SSDs in a liquid cooled system)

Possible Solution
(conceptual)



Call to Action

- We have an opportunity to provide better cooling for SSDs
- How to get involved in the project/sub-project community
  - Reach out to Brad Zakaib (brad.zakaib@coolitsystems.com)
- Where to find additional information (URL links)
  - Project work: https://drive.google.com/drive/folders/1ZK_3wRb4_wA3Fm0_QkKkhlVSjo9tBacM

COOLING:
COLD PLATE

Cold Plate Coolants Workstream Overview

Keegan Yaroch

Associate TS&D Scientist, Dow Chemical

OCP BRONZE™ Member

Cold Plate Coolants - Workstream Details

The OCP Cold Plate Coolants Workstream was created to establish open, vendor-neutral guidance and baseline specifications for coolants used in direct-to-chip liquid cooled systems in order to support safe, reliable and scalable deployment.



Current Focus Areas:

- Water-based coolants
- Propylene glycol-based coolants (PG25)
- Two-phase dielectric coolants
- Fluid quality and monitoring
- Long-term reliability

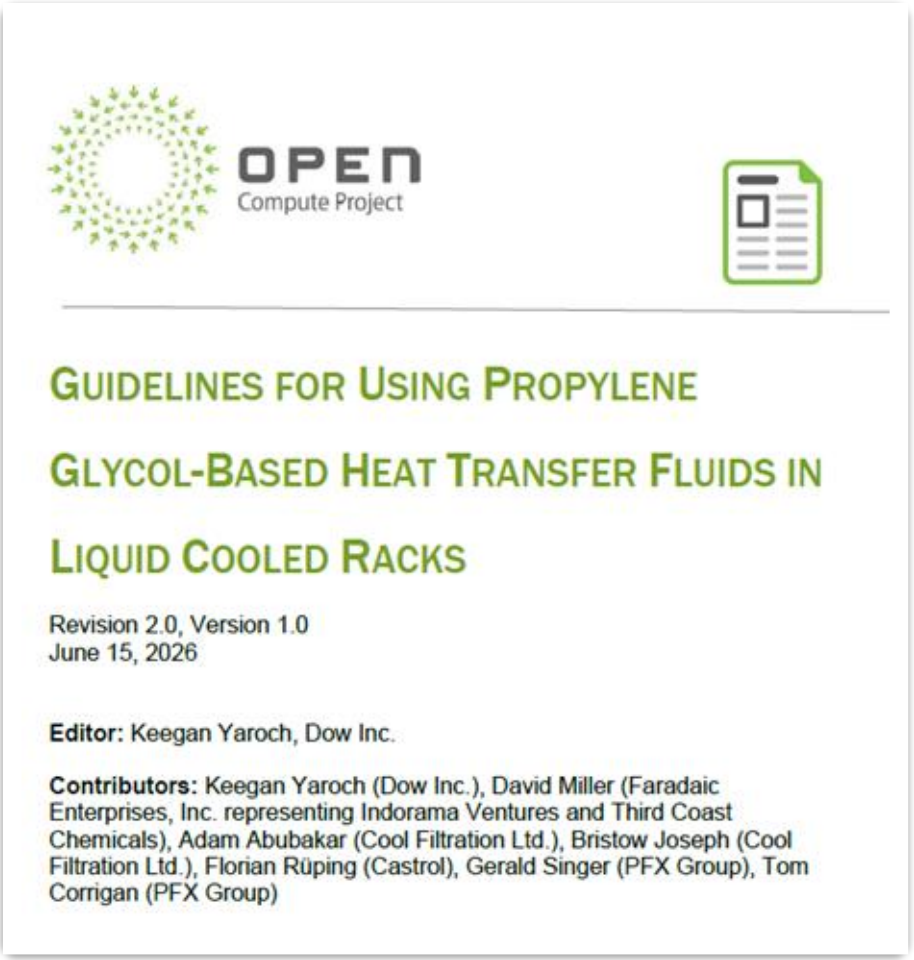
Meetings occur monthly, typically 2nd Monday of the month, from 10-11A EST

2026 Publications – PG-Based Coolants

PG25 (Propylene Glycol-Based Coolants)

- **Base Specification for Single-Phase Propylene Glycol-Based Cold Plate Coolants (PG 25)**
  - ***Editor:*** *Keegan Yaroch (Dow Chemical.),* ***Contributors:*** *Keegan Yaroch (Dow Chemical.), Gerald Singer (PFX Group), Tom Corrigan (PFX Group), Florian Rueping (Castrol), David Miller (Faradaic Enterprises, Inc. representing Indorama Ventures and Third Coast Chemicals)*

- **Guidelines for Using Propylene Glycol-Based Heat Transfer Fluids in Liquid Cooled Racks**
  - ***Editor:*** *Keegan Yaroch, Dow Chemical.,* ***Contributors:*** *Keegan Yaroch (Dow Chemical.), David Miller (Faradaic Enterprises, Inc. representing Indorama Ventures and Third Coast Chemicals), Adam Abubakar (Cool Filtration Ltd.), Bristow Joseph (Cool Filtration Ltd.), Florian Rüping (Castrol), Gerald Singer (PFX Group), Tom Corrigan (PFX Group)*
  - ***This document replaces an earlier version from 2022***



OPEN Compute Project

GUIDELINES FOR USING PROPYLENE GLYCOL-BASED HEAT TRANSFER FLUIDS IN LIQUID COOLED RACKS

Revision 2.0, Version 1.0
June 15, 2026

Editor: Keegan Yaroch, Dow Inc.

Contributors: Keegan Yaroch (Dow Inc.), David Miller (Faradaic Enterprises, Inc. representing Indorama Ventures and Third Coast Chemicals), Adam Abubakar (Cool Filtration Ltd.), Bristow Joseph (Cool Filtration Ltd.), Florian Rüping (Castrol), Gerald Singer (PFX Group), Tom Corrigan (PFX Group)

PG25 Coolants

Guidelines

Purpose: Provides best practices for the deployment and maintenance of propylene glycol-based coolants, primarily intended for coolant users.

Key Topics:

- Material compatibility validation
- Filtration and cleanliness management
- Flushing and commissioning
- Fluid analysis and monitoring
- Inline monitoring strategies
- Corrosion testing
- Fluid lifecycle management

Base Specification

Purpose: Establishes a common baseline for PG25 coolant quality and performance criteria, primarily intended for coolant vendors.

Key Requirements:

- Propylene glycol concentration and purity
- Water quality limits
- Product consistency
- Corrosion protection
- Thermal stability
- Documentation and traceability
- Manufacturing quality controls

2026 Publications – Water-Based Coolants

Water-Based Coolants

- **Base Specification for Single Phase Water-Based Cold Plate Coolants**
  - ***Author:*** *Philip Yu (ChemTreat, Inc.),* ***Contributors:*** *Matt Hatley (Fluid2Chip), Richard Tribble (ChemTreat), Michael Coffin (Individual), Senne Degroote (Arteco Coolants), Gerald Singer (PFX Group), and Florian Rüping (Castrol).*

- **Guidelines for Using Water-Based Heat Transfer Fluid in Single-Phase Cold Plate-Based Liquid-Cooled Racks**
  - ***Editor****: F. Philip Yu (ChemTreat, Inc.),* ***Contributors:*** *Matt Hatley (Fluid2Chip), Michael Coffin (Individual), Senne Degroote and Ronald Merckx (Arteco Coolants), Sean Sivapalan (NVIDIA), Gerald Singer (PFX Group), Eric Yang (Vantage Data Center), Florian Rüping (Castrol), and Dolly Batra (Ecolab).*
  - ***In review process, will be published November 2026. This document replaces an earlier version from 2022.***



Water-Based Coolants

Guidelines

Purpose: Provides best practices for the deployment and maintenance of water-based coolants, primarily intended for coolant users.

Key Topics:

- Water quality requirements
- Corrosion control
- Microbiological control
- Wetted materials recommendations
- Filtration practices
- Commissioning procedures
- Fluid monitoring and maintenance
- Long-term operational guidance

Base Specification

Purpose: Establishes a common baseline for water-based coolant quality and performance criteria, primarily intended for coolant vendors.

Key Topics:

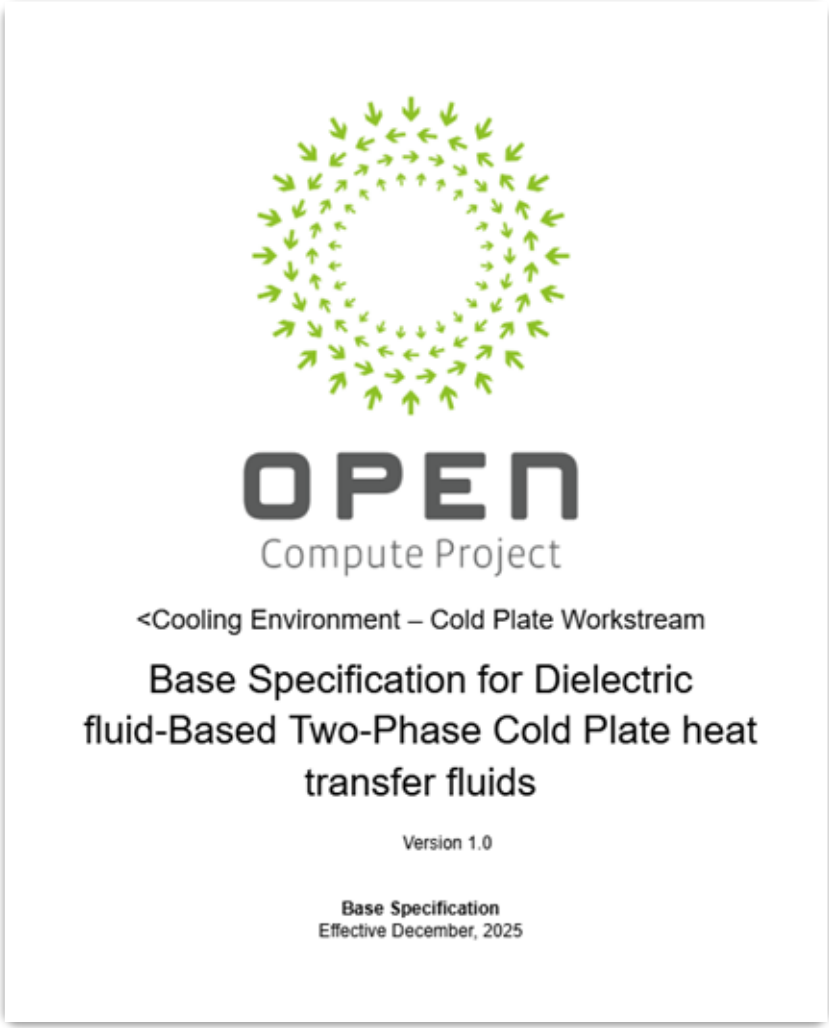
- Fluid and water quality requirements
- Corrosion protection
  - Microbiological protection (preservative)
- Cleanliness requirements

2026 Publications – Dielectric Coolants

Dielectric Coolants – 2-Phase

- **Guidelines for Using Dielectric Heat Transfer Fluids in Two Phase Cold Plate-Based Liquid-Cooled Racks**
  - **Authors:** *Shahar Belkin (ZutaCore), Philip Yu (ChemTreat), Lilach Butchmits (ZutaCore)*

- **Base Specification for Two-Phase Dielectric Fluid-Based Heat Transfer Fluid**
  - **Authors:** Shahar Belkin (ZutaCore), Lilach Butchmits (ZutaCore)



Dielectric Coolants (Two-Phase)

Guidelines

Purpose: Provides best practices for the deployment and maintenance of two-phase dielectric coolants, primarily intended for coolant users.

Key Topics:

- Fluid selection and performance criteria
- Wetted materials compatibility and validation
- System design and filtration requirements
- Installation and commissioning practices
- Monitoring and maintenance
- Leak management & safety considerations
- Fluid lifecycle management

Base Specification

Purpose: Establishes a common baseline for two-phase dielectric coolant quality and performance criteria, primarily intended for coolant vendors.

Key Topics:

- Thermal performance
- Boiling point, density and latent heat requirements
- Fluid purity and cleanliness requirements
- Material compatibility expectations
- Safety and environmental criteria
- Performance validation and testing requirements

Call to Action

- **Get Involved**
  - Join monthly calls and contribute to future initiatives
  - Bring forward new topics of interest
  - Provide feedback on published and newly proposed documents
    - Published Documents: https://www.opencompute.org/contributions/
  - Visit https://www.opencompute.org/community/cold-plate for more information

Guideline Publications

PG	Water	Dielectric
	Expected November 2026	

Base Specification Publications

PG	Water	Dielectric

COOLING:
COLD PLATE

Wetted Materials

Dr. Florian Rüping

Expert Technologist - Global Data Center Platform EG/PG

Castrol

OCP PLATINUM Member

Wetted Materials

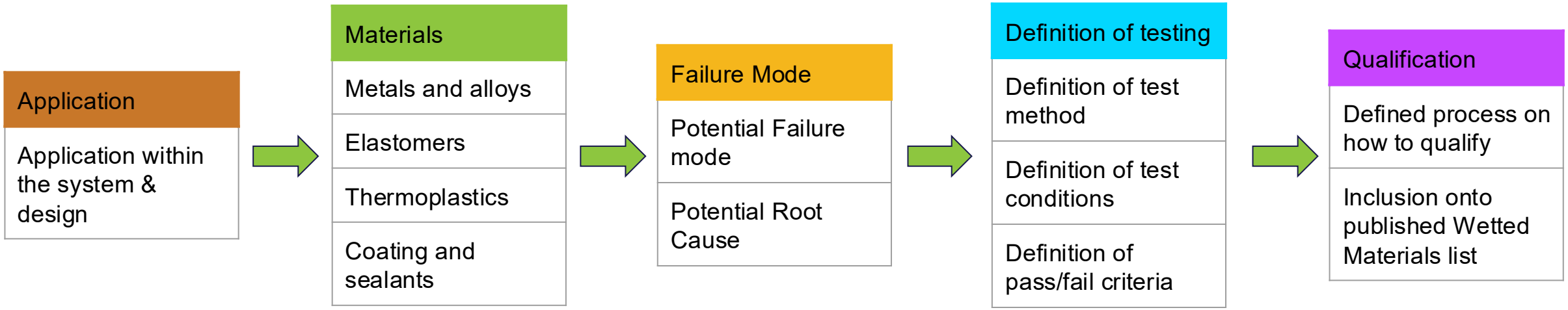
- Goal: The goal of the workstream is to establish a **realistic** approach to approve the compatibility of a coolant with materials. This includes the definition of proper **testing methods**, **conditions** and **pass/fail** criteria. A comprehensive wetted material list shall be established and published as a living document.
- Scope: Coolants used for cold-plate applications including **PG-based**, **Water-based** and **dielectric fluids** with all potentially wetted materials.
- Out of scope: Coolants used for other applications than cold-plates.
- Co-leads: Alex Pica and Florian Rüping





Wetted Materials

- Methodology: The methodology of the workstream focusses on potential Failure Modes and Root Cause(s). Based on these, test methods will be identified to verify that the potential Failure will not occur within the application.



Wetted Materials

Example:

Application
Piping

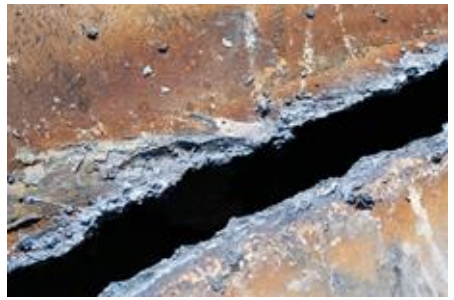
Materials
PP-R
PP-RCT
PP-H

Failure Mode
Environmental Stress Cracking
Embrittlement
Swelling/ Fluid Absorption
Creep

Definition of testing
ASTM D543
60/80 °C, 1008 h
Change of - Weight: <5% - Dimensions: <5% - Tensile strength <10% - Elongation at break: <10% - Observation of Crack initiation: Comparable to the behavior with water

Qualification
Execute testing
Evaluate results based on pass/fail criteria
If positiv: Reach out to update the Approved Wetted Materials List





Wetted Materials

- Challenges:
  - Similar coolants might interact differently with the same material e.g. one PG25 might be compatible while another one might not
  - Materials properties change based on the specific composition. Therefore, testing with one e.g. PA does not mean compatibility is verified with other PA

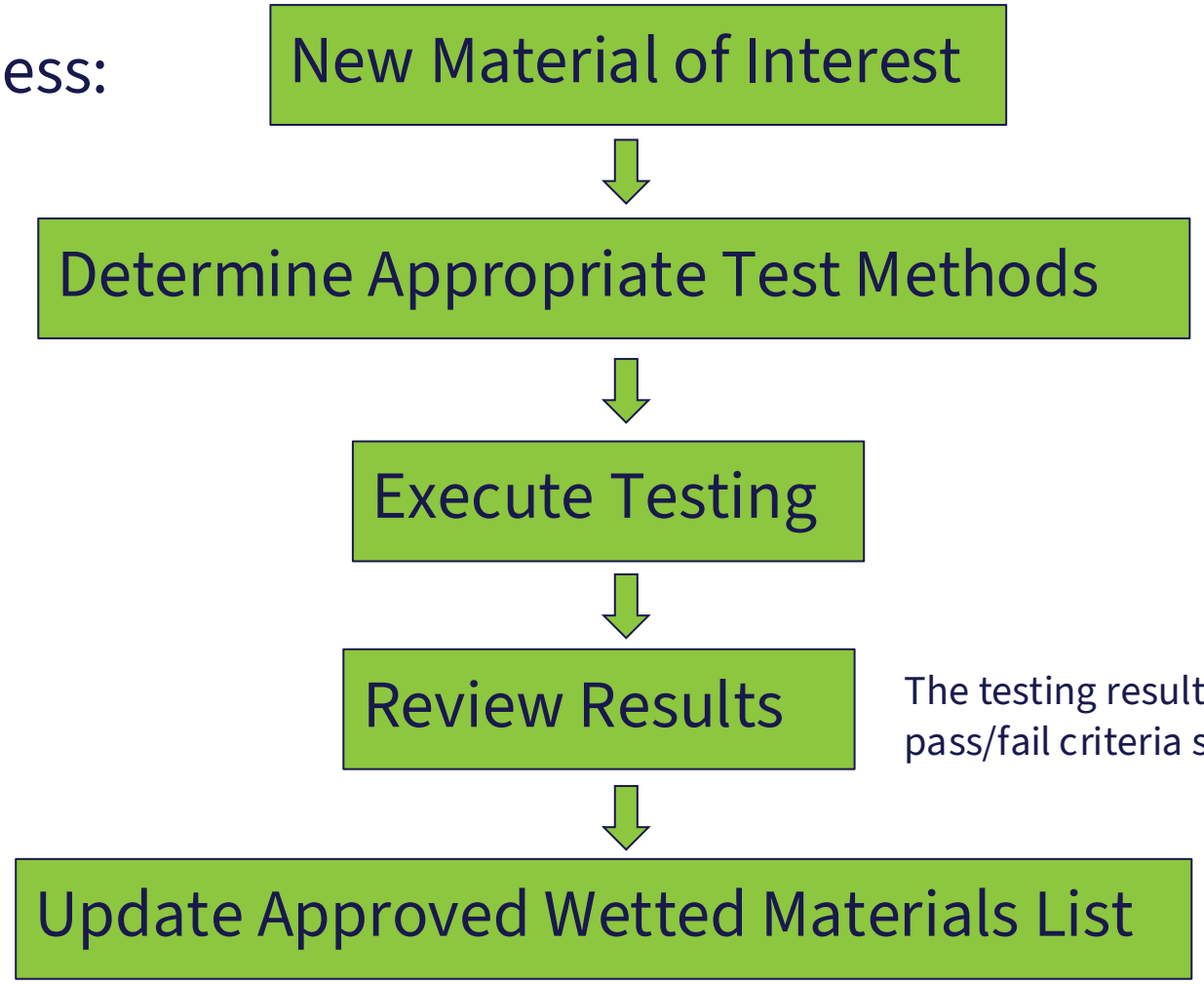




RJC Mold, “The Difference Between PA6 and PA66 Materials,” *RJC Mold*, Dec. 29, 2025. [Online]. Available: The Difference Between PA6 and PA66 Materials. Accessed: Aug. 24, 2026.

Wetted Materials

- Approval process:



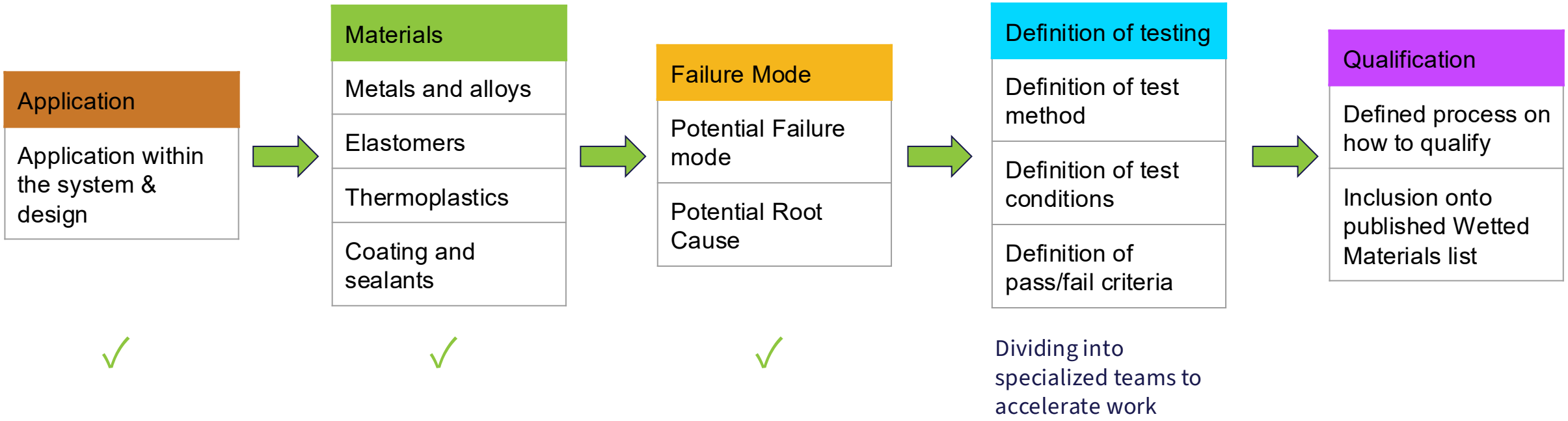
Based on the material's classification and the coolant fluid, the guidance document will specify the testing that must be completed in order to be added to the Approved Wetted Materials List.

The testing results are received and compared to the pass/fail criteria specified in the guidance document.

The Approved Wetted Materials List is updated to include the new material, along with supporting evidence/documentation

Wetted Materials

- Methodology: The methodology of the workstream focusses on potential Failure Modes and Root Cause(s). Based on these, test methods will be identified to verify that the potential Failure will not occur within the application.



Wetted Materials

Failure mode category	Failure mode subcategory 1	Failure mode subcategory 2	Specific measurables/ target
Fluid performance	Fluid contamination	Metals	Iron, Copper, Aluminum, …
		Nonmetals	Chloride, Sulfate, …
		Other	Particulates/ Turbidity
	Fluid component depletion/ leaching		Corrosion inhibitor
			Surfactant/ Antifoam
			Antimicrobial properties
	pH change, acid generation	Degradation	pH, acid value
	Fluid properties		Thermal conductivity, electrical conductivity, viscosity
	Other coolant quality indicators		Refractive index, color, turbidity, particulate count, TOC

Failure mode category	Failure mode subcategory 1	Failure mode subcategory 2	Specific measurables/ target
Component performance	Embrittlement		Impact strength
	Swelling/ Dimensional change		Mass/ Volume/ SpG change
	Environmental stress cracking		
	Other physical property changes	Properties under tension	Ultimate tensile strength
			Ultimate elongation
			Modulus
		Properties under compression	Compression set
			Creep
			Hardness (Durometer)
	Other material degradation indicators		Color change

Call to Action

- Let’s create a reproducible and comparable way of validating material compatibility within the cold-plate environment
- Biweekly workstream sessions every other Wednesday at 9 am CT
- Reach out to Alex Pica (Alex.Pica@saint-gobain.com) or Florian Rüping (florian.rueping@castrol.com)
- ocp-wetted-materials@OCP-All.groups.io | Home



COOLING
ENVIRONMENTS

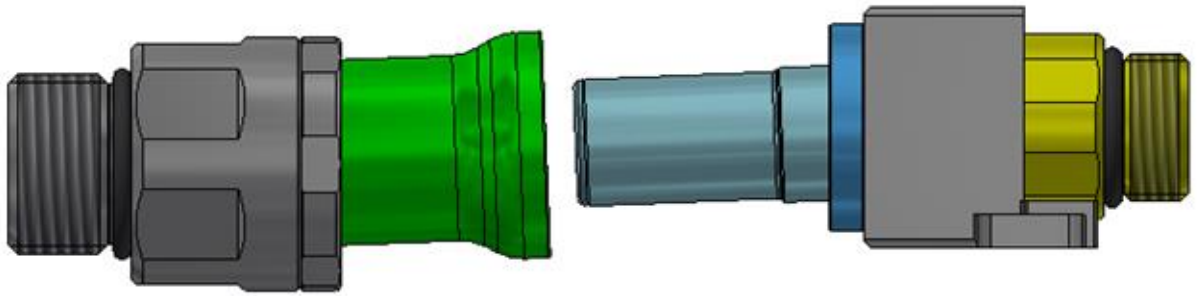
PBMC Specification & Updates

Tim Marquis, Engineering Manager, Parker Hannifin

OCP BRONZE™ Member

Problem Statement

- Develop a multi-sourced interoperable blind mate valve solution to support the ***higher flow*** rates needed for higher power systems.
- Adapt to large valve misalignments common in serviceable IT gear
- Reduced IT gear hose related space constraints



Accommodating Misalignment

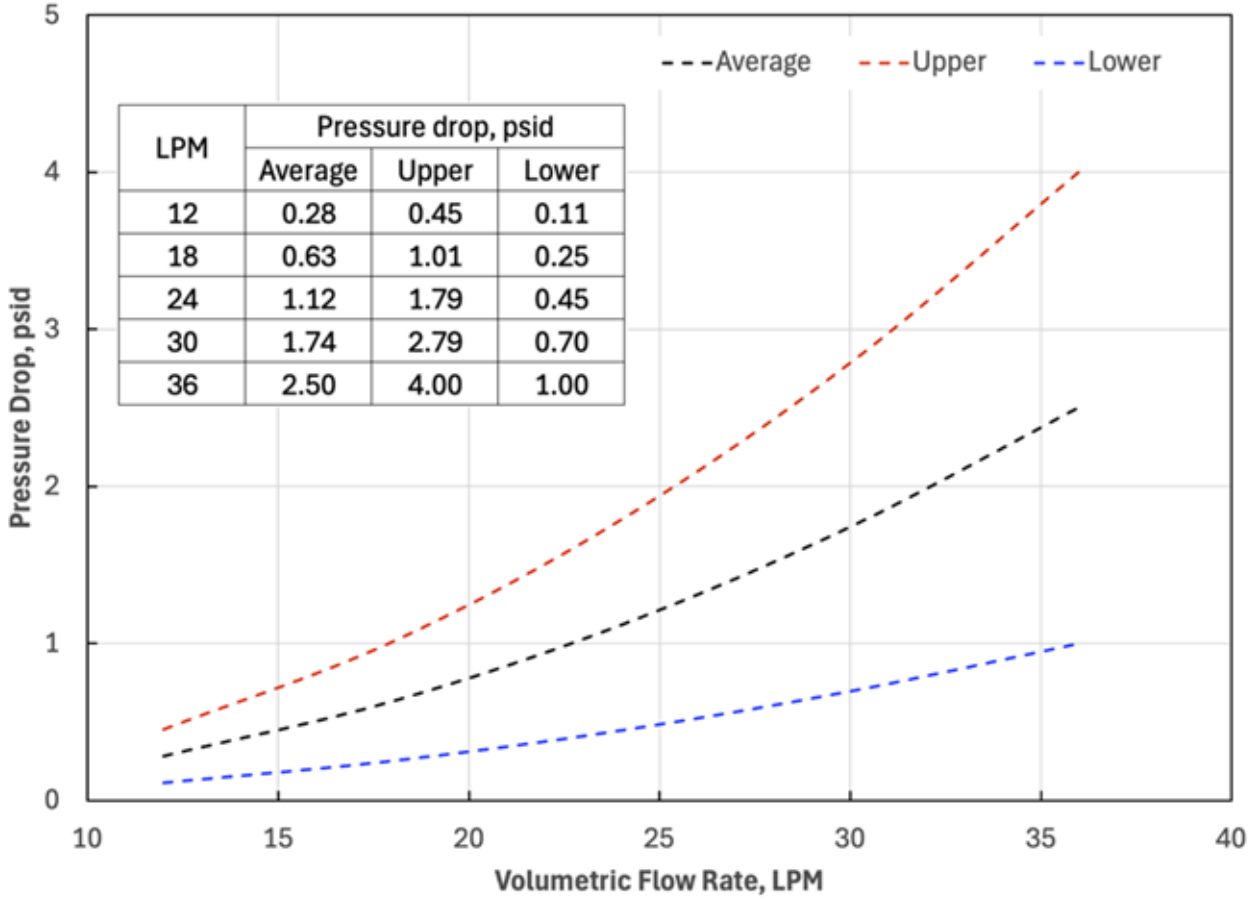
- Unique, all inclusive, pivoting mechanical solution that accommodates large misalignment and mating ranges needed for reliable connections in modular IT gear designs
- Accommodates:
    - 5mm Radial Offset
    - 5mm & 2.5 Deg combined within 5mm radial zone
    - 6.5mm mate distance working range
- Valves self-center when unmated



Design Fundamentals

- High flow & low pressure drop (36 lpm ΔP of <4.0 psi)
- Maximum working pressure of 75 PSIG

LPM	Pressure drop, psid		
	Average	Upper	Lower
12	0.28	0.45	0.11
18	0.63	1.01	0.25
24	1.12	1.79	0.45
30	1.74	2.79	0.70
36	2.50	4.00	1.00



Design Fundamentals

- Supports 1RU/OU form factor
- IT Gear side (plug valve) hose interface is an ORB industry standard
- Innovative plug: Decouples the hose connection from the valve pivot mechanism for the blind mate interface
  - Minimizes the space impact on IT gear design with respect to hose internal routing
  - Enables hose design optionality as the hose stiffness does not affect valve movement during mating
- IT Gear side (plug valve) hose interface has the potential to support various hose connection options
- Full interoperability testing is at the core of the design process.

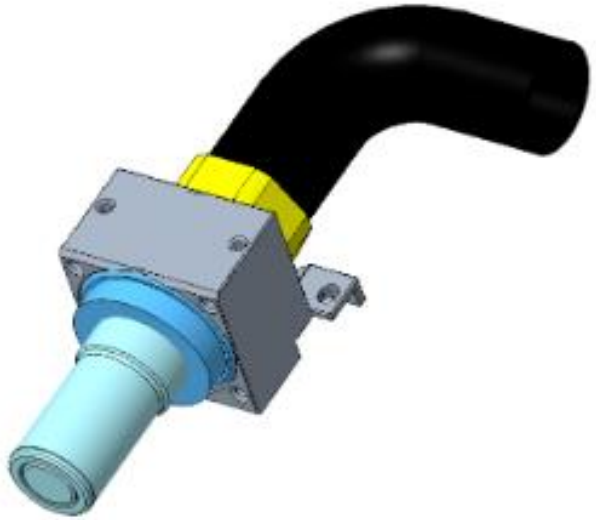
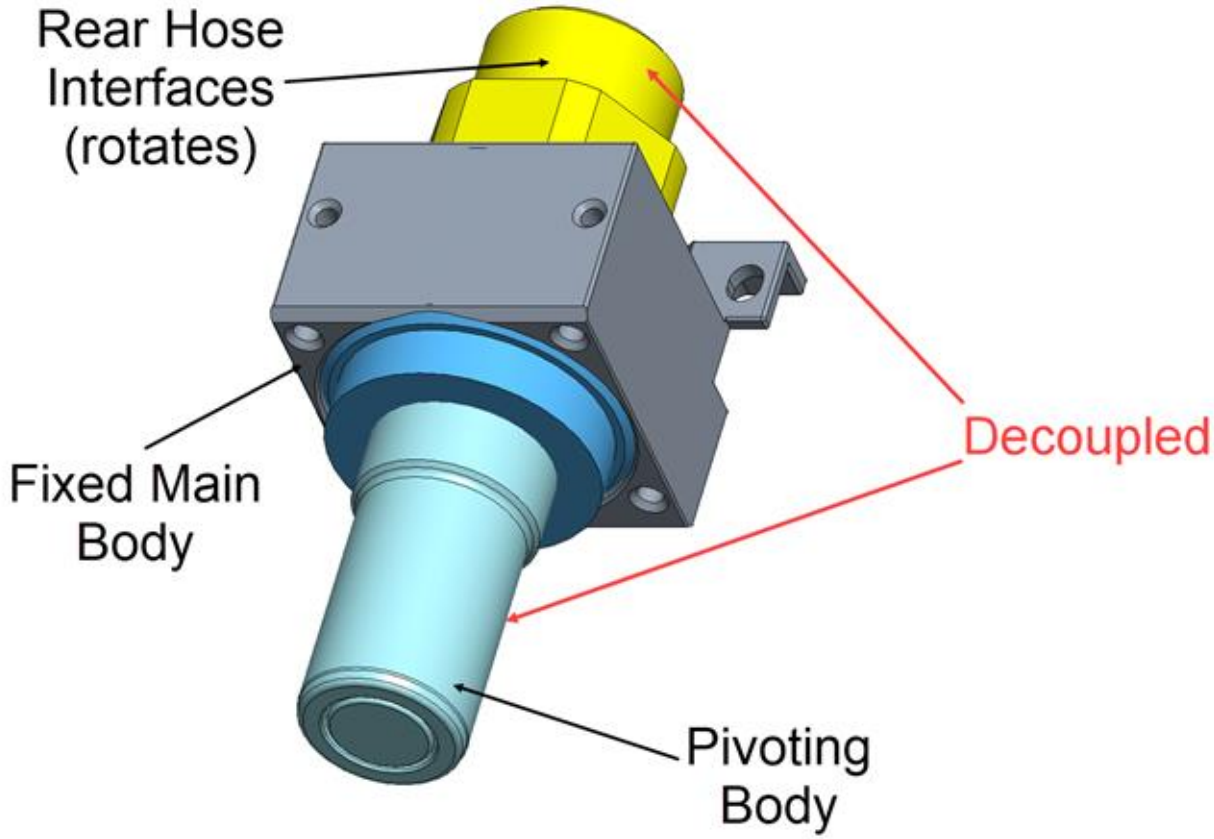


Figure 4.1.2 Plug with hose connected with close bend

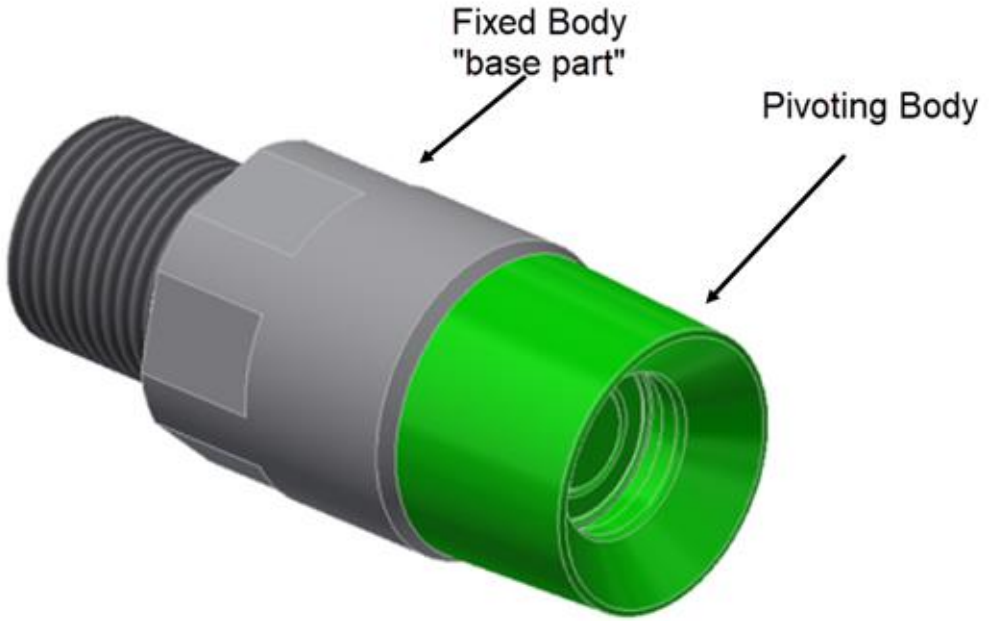
Plug Design

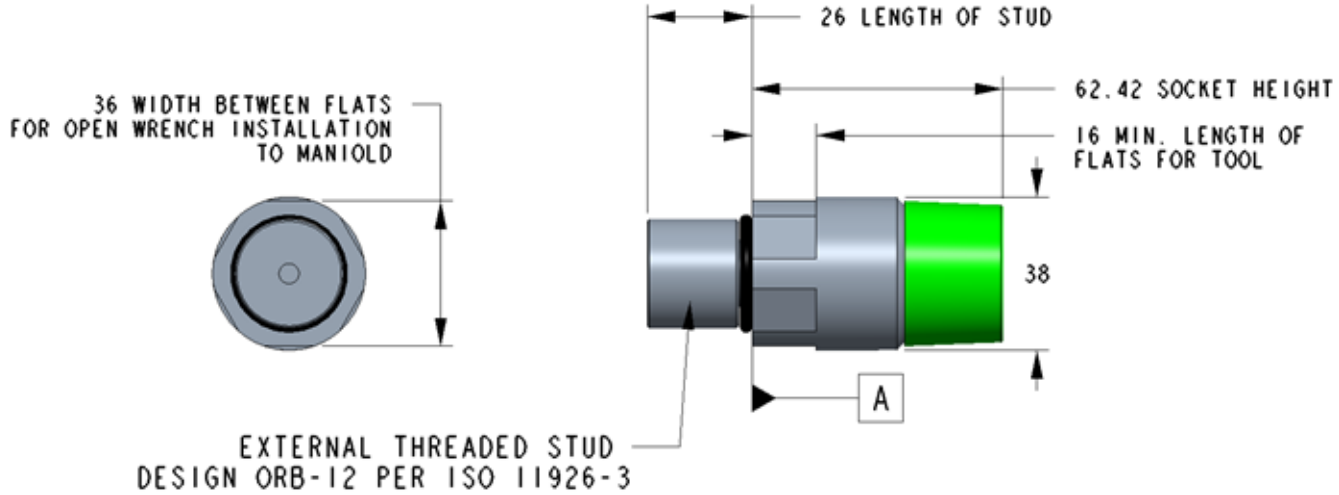
- Multiple mounting interfaces
- ORB hose interface
- Pivoting body adapts to misalignment



Socket Design

- Fixed body with threaded interface and flats for open face wrench
- Pivoting body adapts to misalignment





Specification Updates

Revised Rev 1.1 Specification

- Added ORB hose interface detail & torque on plug valve (was in prelim design guide)
- increased torque in test 2.6.1 A from 6N to 20N, revised figure 2.6.1.A
- New test procedures 2.6.1.B, new Figure 2.6.1.B
- Test 7.1.20
  - Increased torque requirements
  - New test procedure with swivel spinning
  - New socket test with axial force
  - Added Hydrostatic pressure post test
- Minor tolerance correction in Figure 5.2.1.1
- Updated PBMC Test Fixture Guidelines and Procedures with new details

2.6.1.A Plug Test condition A: Swivel Strength Test With Fluid (Includes Pivoting Body Rotation)

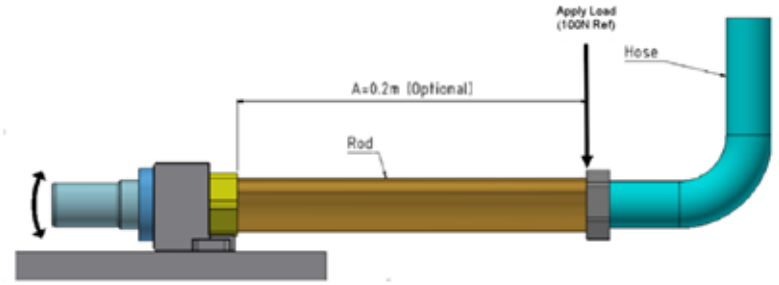


Figure 2.6.1.A Reference Plug Test Set-up

2.6.1B Plug Test condition B (with rotation): Swivel Strength Test without Fluid

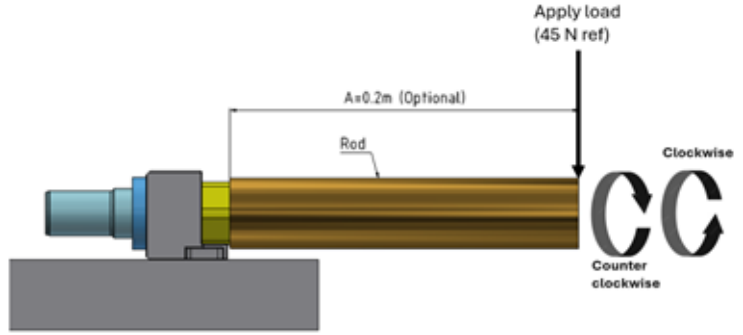


Figure 2.6.1.B Reference Plug Test Set-up

Published

Available on OCP Contribution Database Site

Call to Action

Cold Plate Subproject

Cooling Environments

- Join the monthly Cold Plate Community Call and Mailing List

Cold Plate

UQDv2.0 Update

Elizabeth Langer

CPC , Sr Mgr R&D & Technology

OCP BRONZE Member

UQDv2.0 is officially released!

SPEC, CE, Universal Quick Disconnect (UQD) Specification V2.0.0

Universal Quick Disconnect (UQD) Specification V2.0.0

Contributor: AMD
Family: Cold Plate
Project: Facilities & Physical Infrastructure > Cooling Environments
Release Year: 2026

AMD

CPC® A DOVER COMPANY

Travis

CEJN®

Parker

STÄUBLI

Danfoss

OPEN Compute Project®

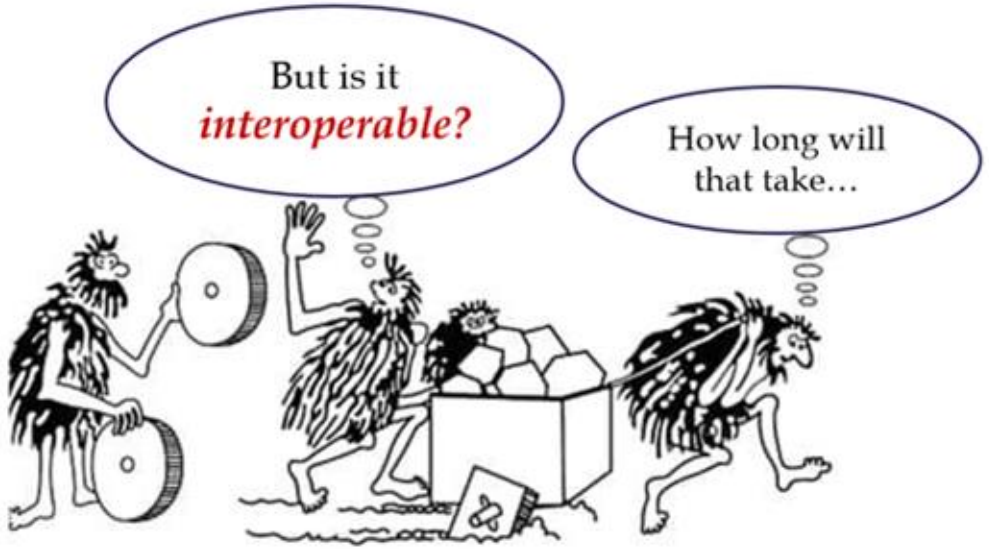
OCP UQD Design Specification V2.0.0

OCP Marketplace - UQD

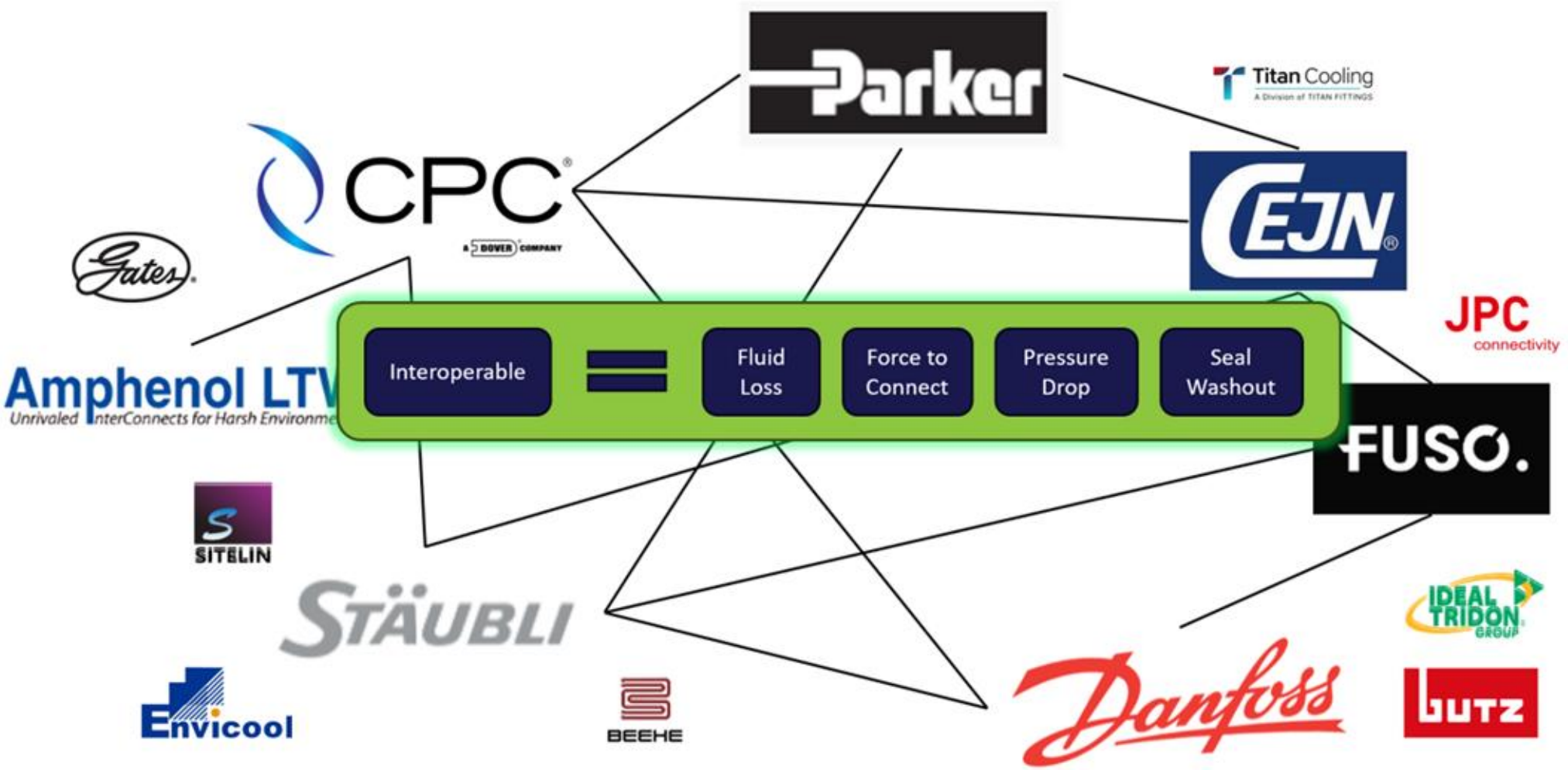
What did we learn?

- We can accomplish A LOT together…we still have much to discuss
- Interoperability is imperative…but tricky in practice…

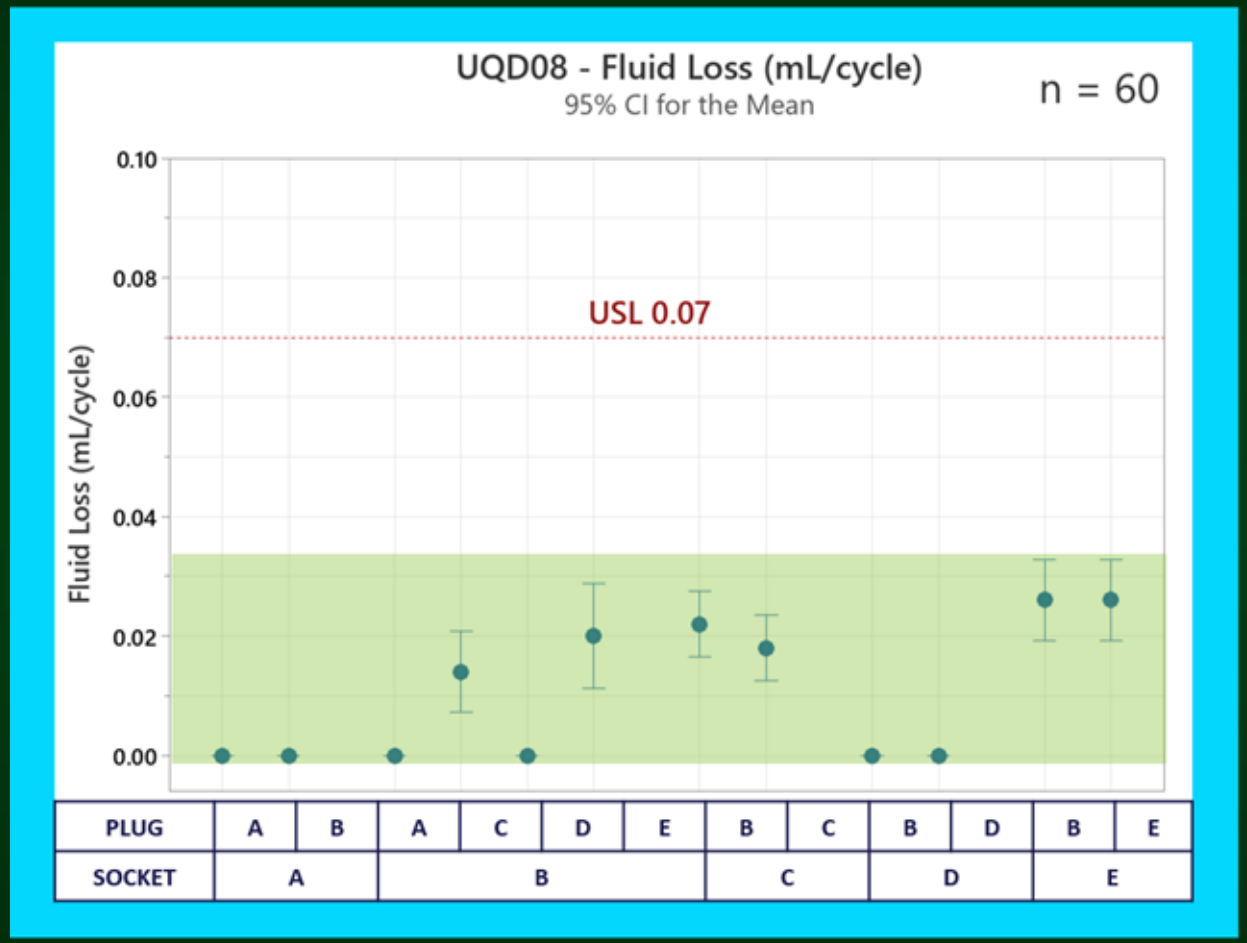
So how did it go for UQDv2.0?



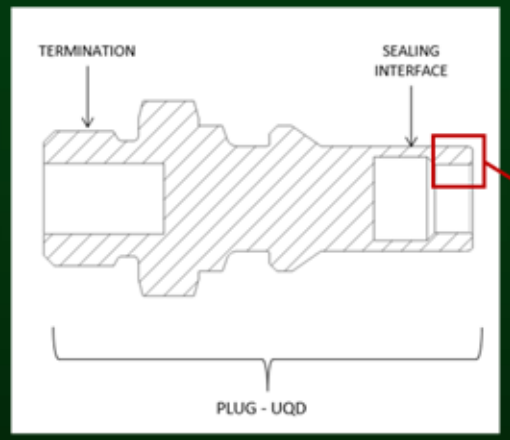
UQDv2.0 pilots interoperability testing!

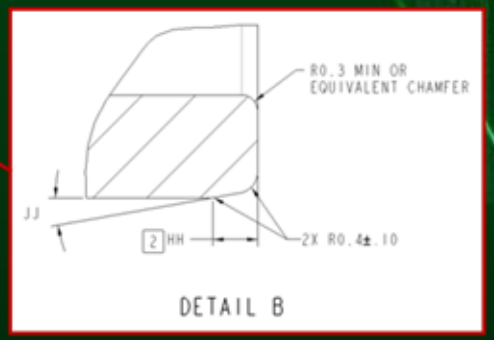


Interoperable = **Fluid Loss** | Force to Connect | Pressure Drop | Seal Washout

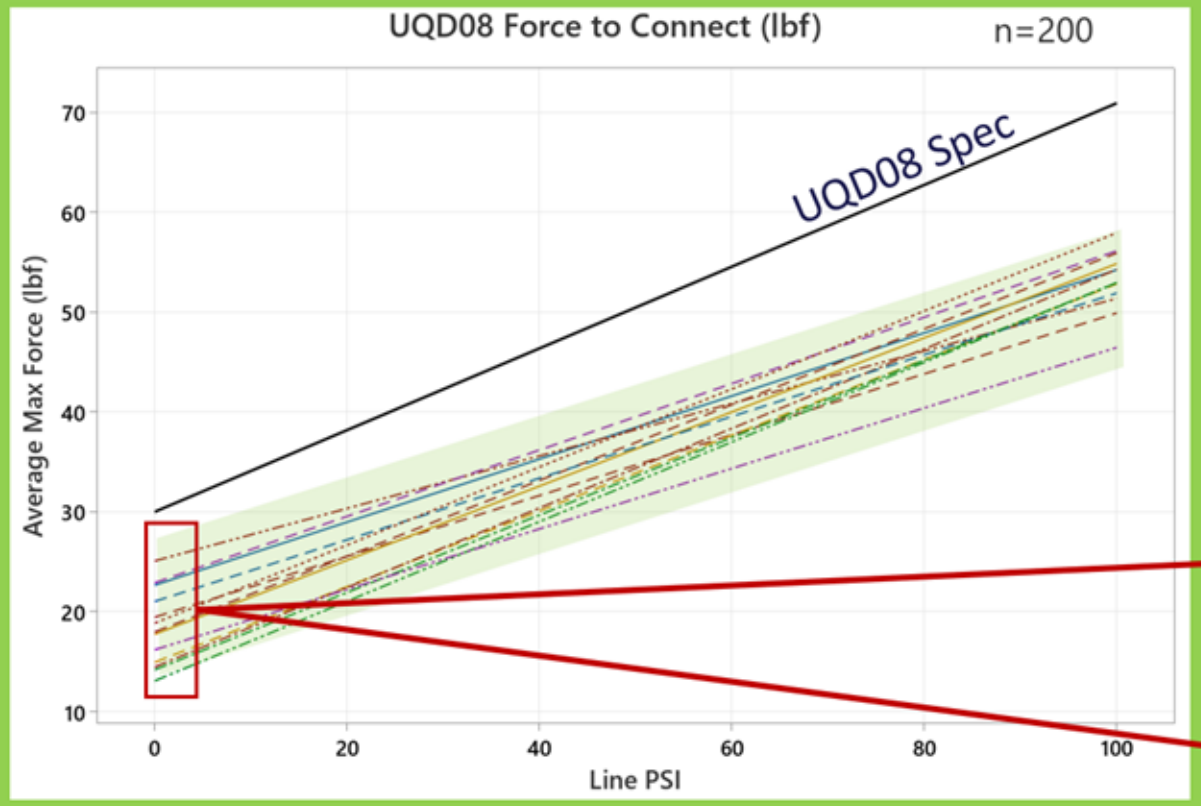


- Standardization of plug chamfer reduces fluid loss variation across suppliers
- *With form/fit defined in specification – should fluid loss still be an interop requirement?*

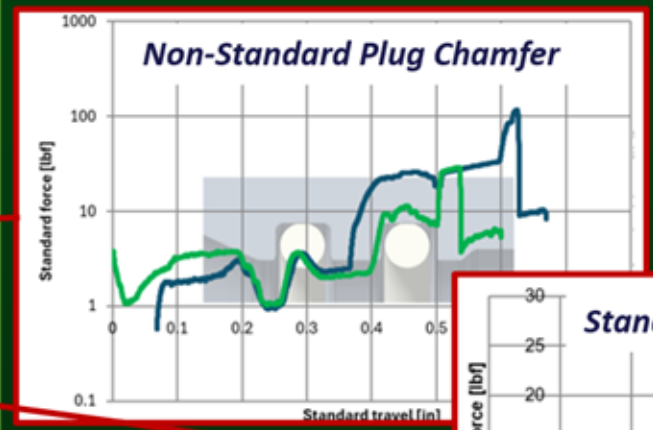


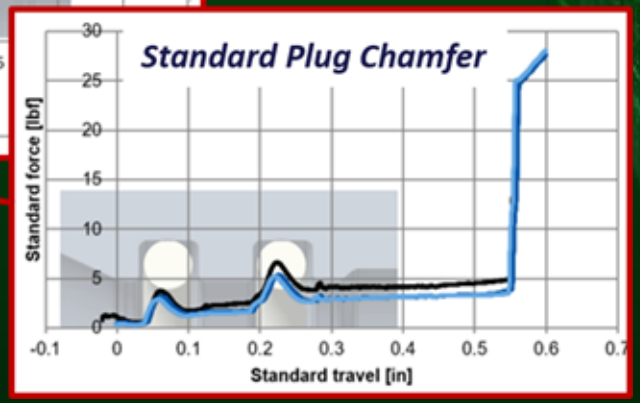


Interoperable = Fluid Loss | **Force to Connect** | Pressure Drop | Seal Washout

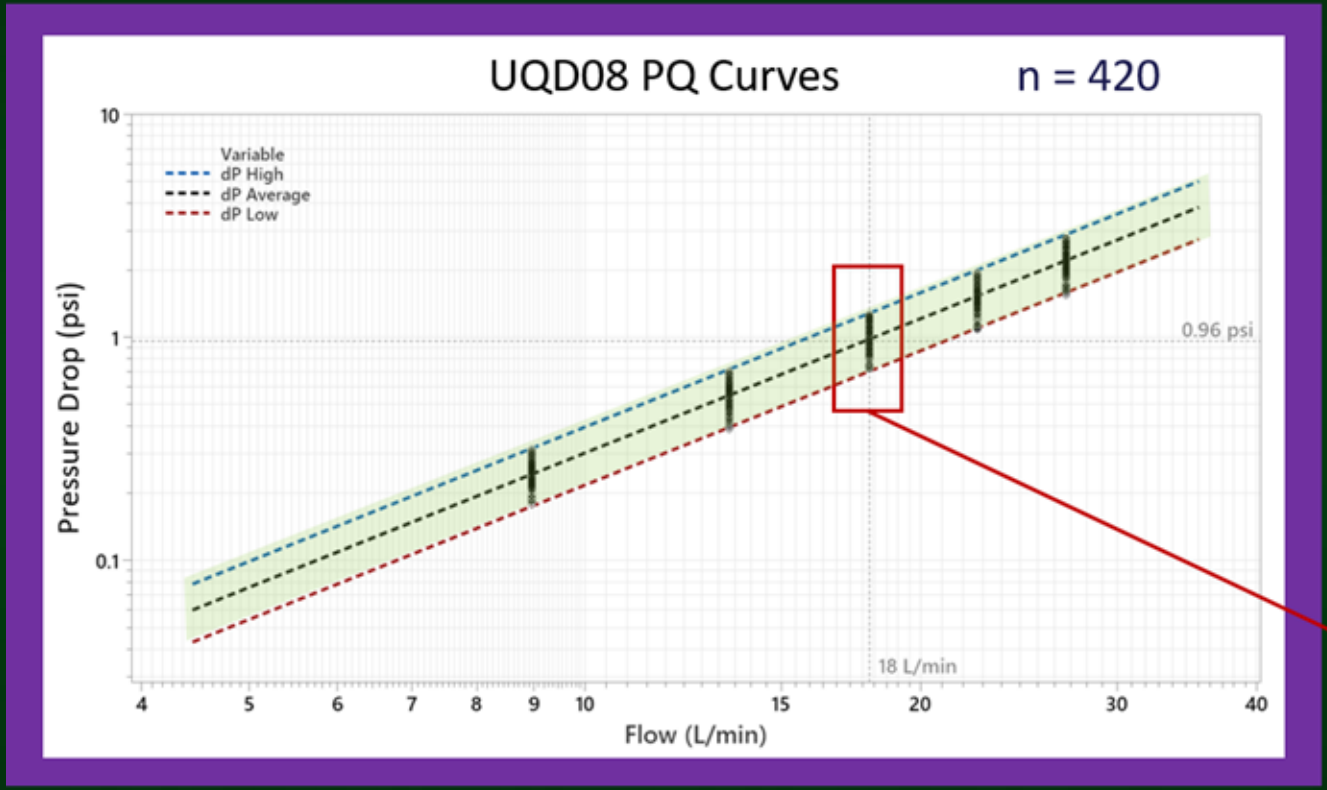


- Characterization beyond 0psig, spec scales with pressure (F=PA)
- *With lead-in chamfer and maximum spring rate defined – should force to connect still be an interop requirement?*

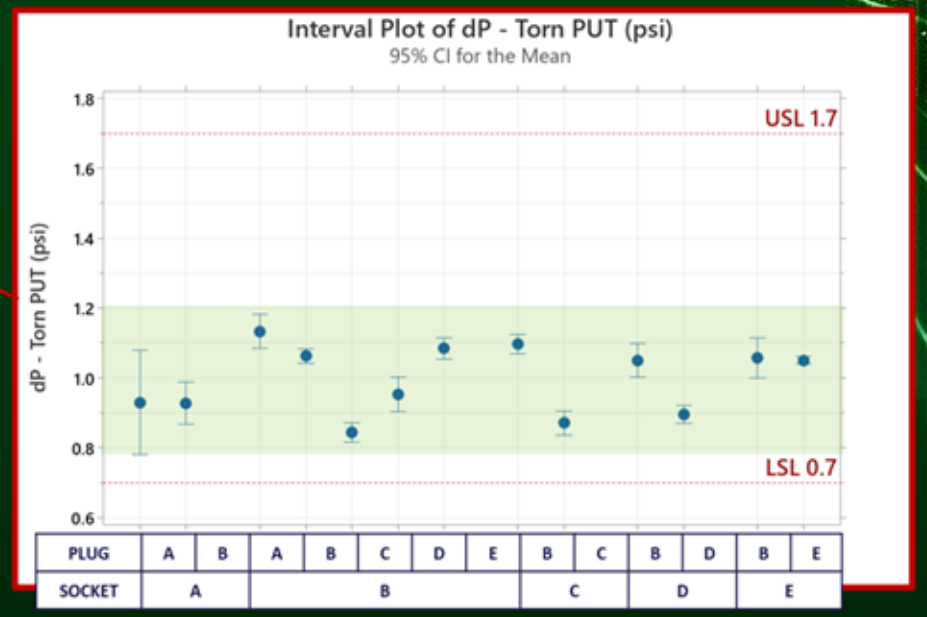




Interoperable = Fluid Loss | Force to Connect | **Pressure Drop** | Seal Washout



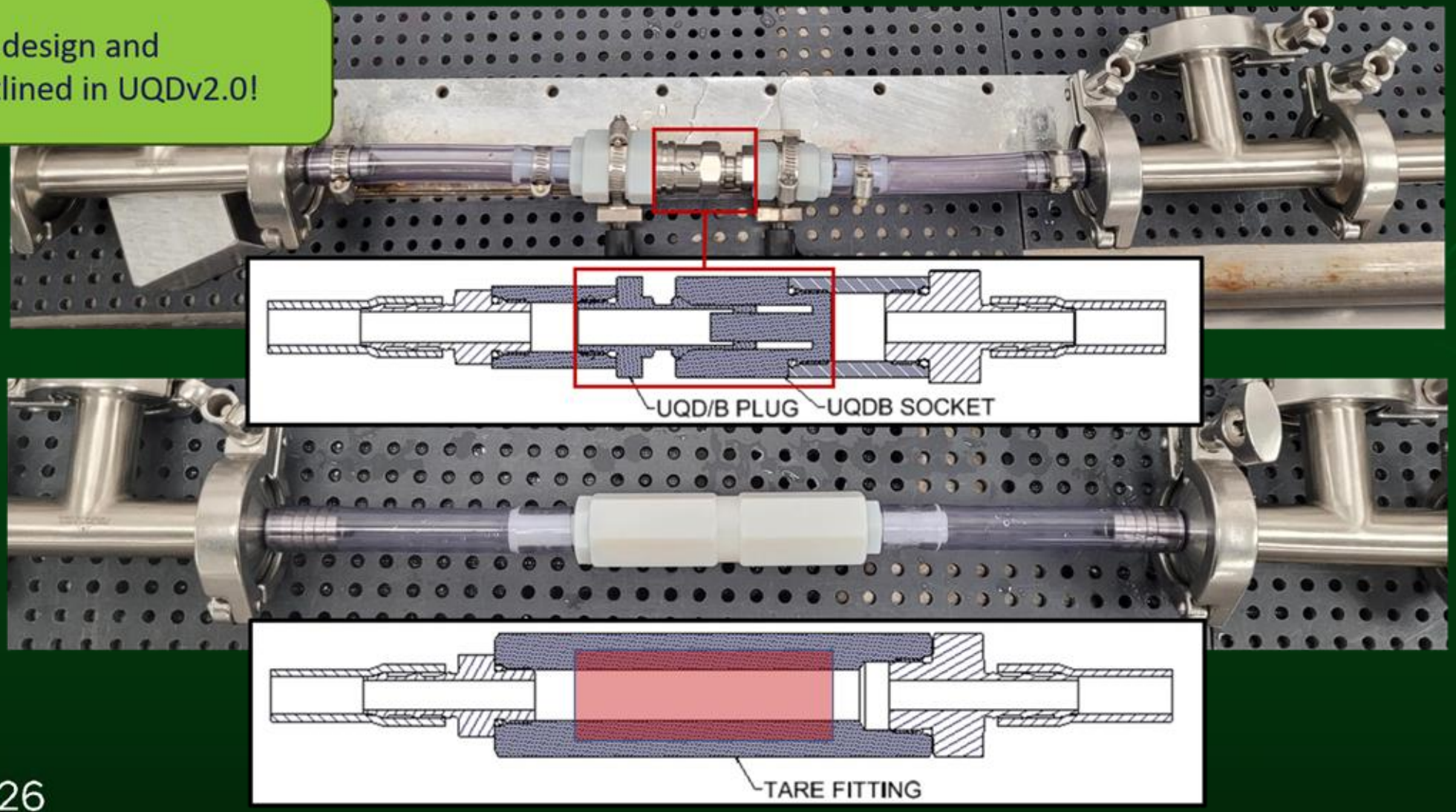
- Use LSL & USL to characterize flow performance (Cv)
- Good consistency across suppliers and flow direction: dP @ Rated Flow = 1.0 ± 0.2 psi
- *What other variation remains to capture with interop dP testing…?*



At a minimum, consistent tare methods

Pressure Drop

Tare fixture design and method outlined in UQDv2.0!



Interoperable = Fluid Loss | Force to Connect | Pressure Drop | **Seal Washout**

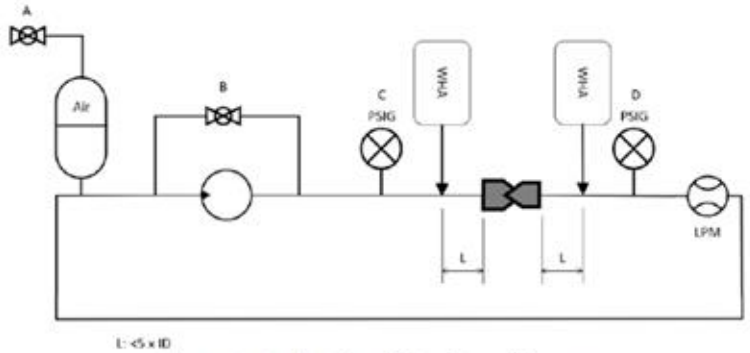


Figure 14 - Seal Washout (Closed Loop) Diagram

Number of Cycles	20, each flow direction *(plug to socket / socket to plug)*	**Connection Speed**	40mm/s ±10mm/s
Test Flowrate	2x the rated flow for the QD	**Test Temperature**	60°C
Test Pressure	6.9 bar(g) (100 psi(g)) operating (Excludes transient pressure spike).	**Test Fluid**	Water or PG25
Acceptance Criteria	No observed leakage or seal washout		

- New test added to UQDv2!
- Directional performance differences observed across supplier sites
- *What is the "correct" way to test for this relative to the application? Multiple conflicting specs…*
- *Is it an interoperability issue or something else…?*



Generic Seal Washout Examples

Interoperability across the ecosystem?

BMQC (2023)



UQDv2 (2026)



MQD (WIP 2026)





MiniQD (WIP 2027)



PBMC (2026)

Opportunity to harmonize interoperability framework and process across QD specifications, best practices and use-cases

Enabling efficient interoperability for the future

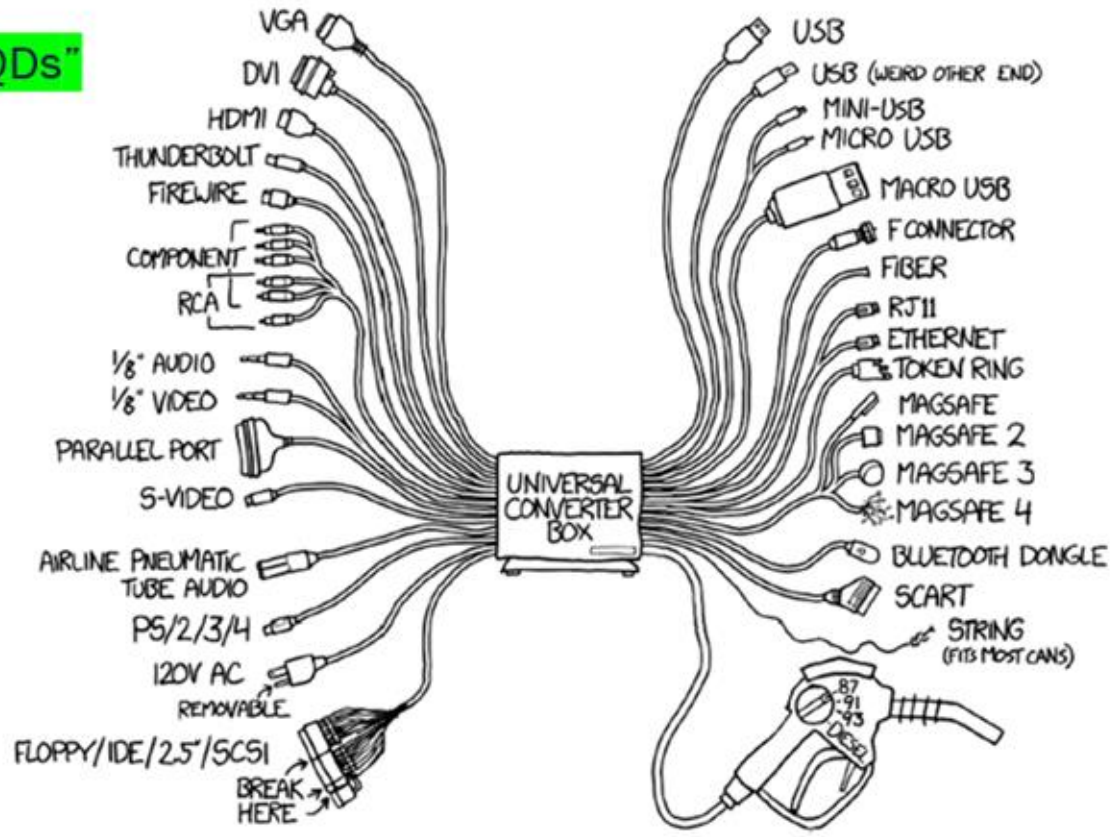
New contribution WIP! *(see call to action next slide)*

"Fundamentals of Verification, Reliability, and Interoperability of QDs"

- *Definition of interoperability for a QD*
- *Different context for QDs based on location in rack*
- *Standard datums / refs to DOF*
- *Tolerance/resolution guidance on key POIs*
- *Testing fundamentals*
- *Interface/interop risk assessments*
- *Context for QDs across integration levels*

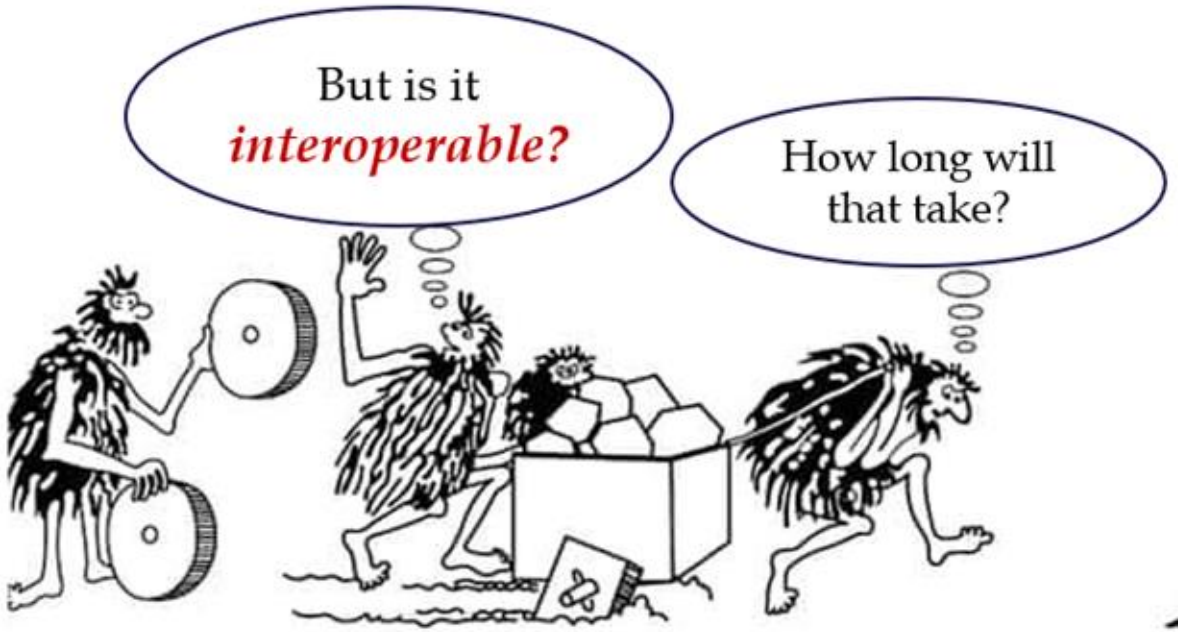
What factors can influence interoperability?

- Form – profile, geometry, interfaces (ie latching)
  - Lot to lot, vendor to vendor dimensional variation
- Valving & Internal Architecture (seal size, mate length)
  - Unexpected interactions across vendors subsystems
- Operational Dynamics and Ratings
  - All specs should have a tolerance/window defined
- Material & Environmental (compatibility)



Universal Converter Box by Randall Munroe, xkcd 2025

Shifting the conversation back to innovation



"QD QED" | proof of interoperability – drives efficiency and allows us to scale



Call to Action - Join the Conversation!

- Explore QDs available in the OCP Marketplace
- Help define the next generation of QDs
  - OCP Cold Plate Project monthly calls/discussion forum
  - Active workstreams accessible on wikisite
  - Enable interoperability of the supply chain
  - Specific to QD base verification document:
    - Connect with Elizabeth Langer - elizabeth.langer@colder.com



Monthly Community Call
(2nd Wednesday of each month)

- Join the mailing lists today:
  - Cooling Environments
  - Cold Plate Subproject

Thank You!

